

ICT ASSIGNMENT 1

5). Decimal to binary:-

$$\bullet (156)_{10} \rightarrow \begin{array}{r} 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ (10011100)_2 \end{array}$$

$$\bullet (1024)_{10} \rightarrow \begin{array}{r} 1024 \ 512 \ 256 \ 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ (100000000000)_2 \end{array}$$

$$\bullet (255)_{10} \rightarrow \begin{array}{r} 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ (11111111)_2 \end{array}$$

Decimal to octal:-

$$\bullet (156)_{10} \rightarrow \begin{array}{r} 156 \div 8 \quad 19 \text{ R } 0.5 \quad 4 \\ 19 \div 8 \quad 2 \text{ R } 0.375 \quad 3 \\ 2 \div 8 \quad 0 \text{ R } 0.25 \quad 2 \end{array} \quad (234)_8$$

$$\bullet (1024)_{10} \rightarrow \begin{array}{r} 1024 \div 8 \quad 128 \text{ R } 0 \quad 0 \\ 128 \div 8 \quad 16 \text{ R } 0 \quad 0 \\ 16 \div 8 \quad 2 \text{ R } 0 \quad 0 \\ 2 \div 8 \quad 0 \text{ R } 0.25 \quad 2 \end{array} \quad (2000)_8$$

$$\bullet (256)_{10} \rightarrow \begin{array}{r} 256 \div 8 \quad 31 \text{ R } 0.875 \quad 7 \\ 31 \div 8 \quad 3 \text{ R } 0.875 \quad 7 \\ 3 \div 8 \quad 0 \text{ R } 0.375 \quad 3 \end{array} \quad (377)_8$$

Decimal to Hexadecimal:-

$$\bullet (156)_{10} \rightarrow \begin{array}{r} 156 \div 16 \quad 9 \text{ R } 0.75 \quad 12 \\ 9 \div 16 \quad 0 \text{ R } 0.5625 \quad 9 \end{array} \quad (9C)_{16}$$

$$\bullet (1024)_{10} \rightarrow \begin{array}{r} 1024 \div 16 \quad 64 \text{ R } 0 \quad 0 \\ 64 \div 16 \quad 4 \text{ R } 0 \quad 0 \\ 4 \div 16 \quad 0 \text{ R } 0.25 \quad 4 \end{array} \quad (400)_{16}$$

GENIUS

$$\begin{array}{rcl}
 (A55)_{10} \rightarrow & 255 \div 16 & 15 \text{ R } 0.9375 & 15 \\
 & 15 \div 16 & 0 \text{ R } 0.9375 & 15 \\
 & & & (FF)_{16}
 \end{array}$$

BINARY ADDITION, BINARY SUBTRACTION:-

1). Binary addition:-

$$\begin{array}{r}
 \begin{array}{ccccccc}
 & 1 & 1 & 1 & 1 & & \\
 1 & 0 & 1 & 1 & 0 & 1 & \\
 + & 1 & 1 & 0 & 1 & 1 & \\
 \hline
 1 & 0 & 0 & 1 & 0 & 0 &
 \end{array}
 \end{array}$$

2). Binary subtraction:-

$$\begin{array}{r}
 111001 \\
 - 1001 \\
 \hline
 110000
 \end{array}$$

Explanation to convert fractional binary number to its decimal equivalent:-

To convert fractional binary number to decimal separate both decimal before part and after one.

like as given in question Ex: 10.101

→ Before decimal one → 110

→ Take out binary part 6

→ After decimal part → 101
 $\downarrow \quad \downarrow$
 $2^{-1} \quad 2^{-3}$

Note:-

- After decimal part has a pattern

$2^{-1}, 2^{-2}, 2^{-3}, 2^{-4}, 2^{-5}, 2^{-6}$

→ After decimal = $2^{-1} + 2^{-3} = 0.625$ → Now just add both parts

$$= 6 + 0.625 \rightarrow \boxed{6.625}$$